Accumulator Monitoring System Integration

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Introduction

The Accumulator Monitoring System (AMS) HMI was developed to monitor and display real-time battery information received from the Battery Management System (BMS) over the CAN bus. An Adafruit Feather M4 CAN microcontroller acts as the interface between the BMS (or CAN Bus Analyser during simulation) and the Nextion display. The system provides visualisation of key battery parameters, warning indicators, and critical fault notifications to assist in monitoring accumulator health and system safety.

The functions within the code were reorganised, the code was commented, and the HMI layout was modified to enable seamless integration between the Arduino firmware and the Nextion display. The system was tested using various CAN Bus Analyser messages to verify its functionality.

Components in the System

Accumulator Final System

Battery Management System → Adafruit M4 CAN → Nextion Display

Simulation Test System

Microchip CAN Bus Analyser → Adafruit M4 CAN → Nextion Display

Power Supply to components

- CAN Bus Analyser (Male Mini-B)-Computer to can analyser
- Adafruit CAN M4 (Male USB-C)
- Nextion Display (Male Micro-B)-Using Male Micro-B adapter socket

Miscellaneous Components

- Male to Male Jumpers
- Male to Female Jumpers
- Male Micro-B Socket
- SD Card (For HMI TFT File)
- Solder and Soldering Iron

Simulation System Pin Connections

1.Micro Can Bus Analyser to Adafruit CAN

Connect Pin 2 (Red), Pin 6 (Yellow), and Pin 7 (Blue) of the CAN Bus Analyzer to CAN_L (Red), GND (Yellow), and CAN_H (Blue) on the Adafruit M4 CAN, respectively. Refer to the images below.

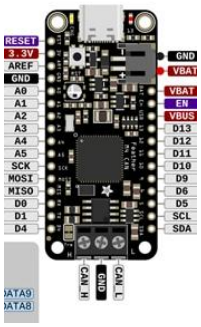


FIGURE 2-1: PINOUT FOR THE DB9 CONNECTOR

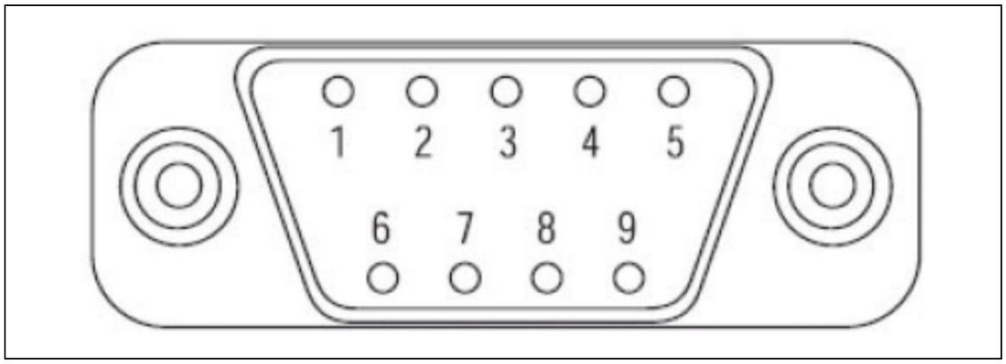


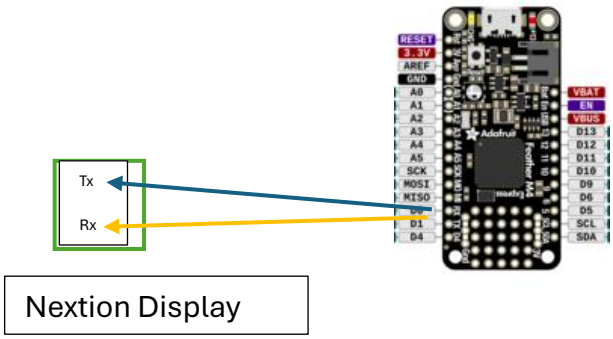
TABLE 2-1: 9-PIN (MALE) D-SUB CAN BUS PINOUT

Pin Number	Signal Name	Signal Description
1	No Connect	N/A
2	CAN_L	Dominant Low
3	GND	Ground
4	No Connect	N/A
5	No Connect	N/A
6	GND	Ground
7	CAN_H	Dominant High
8	No Connect	N/A
9	No Connect	N/A

2.Adafruit CAN to Nextion Display

The Rx terminal (Blue) from Adafruit M4 Can to Tx (Blue)of Nextion Display.

The Tx terminal (Yellow) from Adafruit M4 Can to Rx (Yellow)of Nextion Display.



Software Required

- Microchip CAN Bus Analyser- https://www.microchip.com/en-us/development-tool/apgdt002
- Arduino IDE- https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE
- Nextion HMI Editor- https://nextion.tech/nextion-editor/#_section1

Setup Guide for Adafruit M4 CAN

Download the arduino IDE

File → **Preferences** → find the "Additional boards manager URLs" box → paste this in:

https://adafruit.github.io/arduino-board-index/package_adafruit_index.json

Tools → **Board** → **Boards Manager** → search **"Adafruit SAMD"** → click Install

Once that installs (takes a minute), go back to Tools → Board and you should now see an **Adafruit SAMD** section with the Feather M4 CAN in it. (SAME51)

Now install the libraries.

Tools → **Manage Libraries**

Search and install these one by one:

1. Search **Adafruit CAN by Adafruit**→ install
2. Search **Easy Nextion Library by Athanasios**→ install

Plug the board in

Tools → **Port** —click the COM3

Guide to upload HMI file to Nextion Display

- Generate a TFT file using the Nextion Editor.
  Go to **File → Generate TFT Output**.
- Copy the generated TFT file to the SD card. Ensure the SD card contains only the TFT file.
- Insert the SD card into the Nextion SD card slot.
- Once the update succeeds, disconnect the power supply and remove the SD card.
- Reconnect the power supply.

Test Cases Completed

Displayed information includes:

- SOC percentage
- Pack voltage and current
- Average cell temperature
- Highest cell temperature
- Hottest cell ID

Warnings are triggered when thresholds are exceeded.

Examples:

Warning	Trigger
Low Battery	SOC < 15%
High Temp	T ≥ 50°C
High Voltage	V ≥ 590V
High Current	I ≥ 450A

Status of Charge (SOC)

7.3.2 SOC Message

SOC is transmitted using a separate CAN message.

Parameter	Value
CAN ID	0x100
Transmission Rate	10 Hz

Signal layout:

Byte	Signal
0-1	SOC × 10

SOC (%)	Condition	Expected HMI Response
SOC > 15%	Normal Operation	SYSTEM OK
5% < SOC ≤ 15%	Low Battery Warning	WARNING
SOC ≤ 5%	Critical Battery	SHUTDOWN

Battery Condition	SOC	Data	Expected HMI	Received HMI	Comments
Full Charge	100%	E8 03	SYSTEM OK, SOC = 100%, No warnings	SYSTEM OK, SOC = 100%, No warnings	
Normal Operation	80%	20 03	SYSTEM OK, SOC = 80%, No warnings	SYSTEM OK, SOC = 80%, No warnings	
Low Battery Warning	15%	96 00	WARNING, Low Battery icon flashing	WARNING, Low Battery icon flashing	
Critical Battery	5%	32 00	SHUTDOWN – STOP CHARGING, CRITICAL FAULT	To be checked again	To be checked again

Thermal Cell Readings Test Conditions

The BMS thermal cell readings message used by the Badenia is transmitted with:

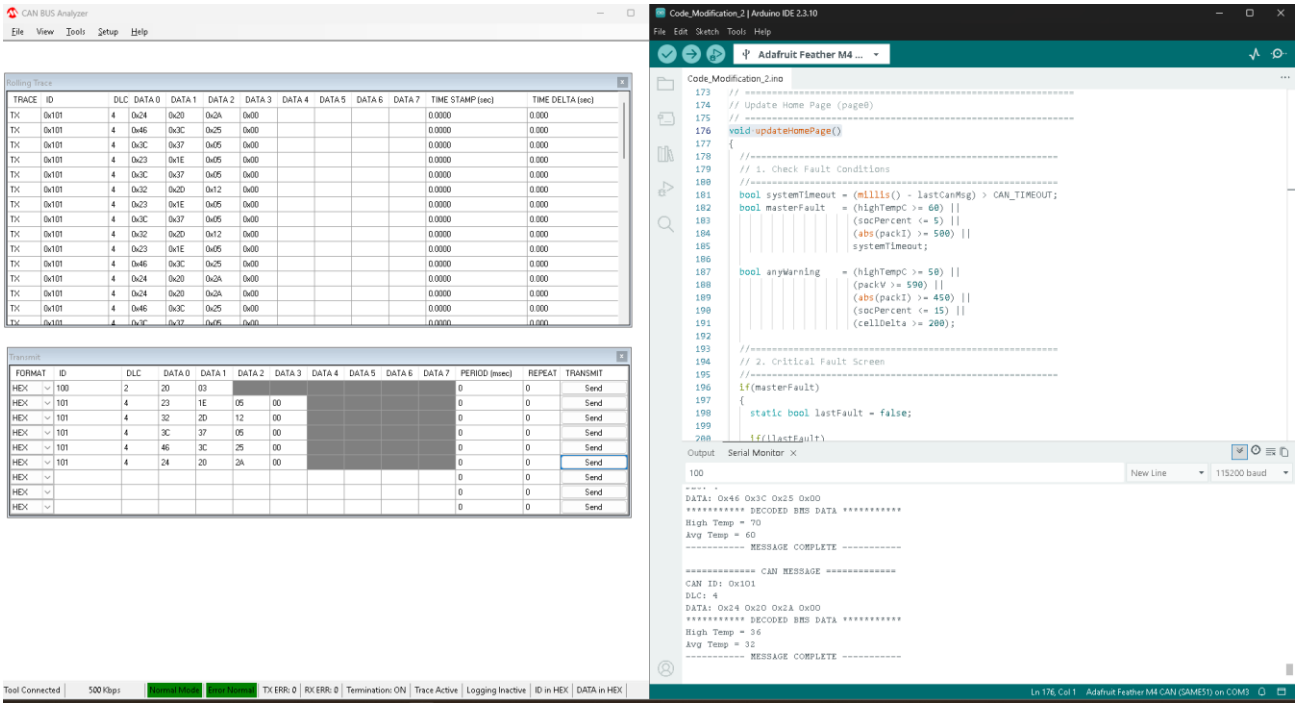
Parameter	Value
CAN ID	0x101
Transmission Rate	10 Hz
Data Length	4 bytes

Signal layout:

Byte	Bits	Signal	Description
0	0-7	Highest Cell Temp	Maximum cell temperature
1	8-15	Average Cell Temp	Mean pack temperature
2	16-23	Hottest Cell ID (LSB)	Cell index
3	24-31	Hottest Cell ID (MSB)	Cell index

Highest Cell Temperature (°C)	Condition	Expected HMI Response
T < 50°C	Normal Operation	SYSTEM OK
50°C ≤ T < 60°C	High Temperature Warning	WARNING
T ≥ 60°C	Critical Temperature	SHUTDOWN

Thermal Cell Condition	Data	Decode Values	Expected HMI		Received HMI	Comments
Normal Operation		23 1E 05 00	Max Temp=35°C Avg Temp=30°C Hottest Cell=5	SYSTEM OK No warnings	SYSTEM OK No warnings Max Temp=35°C Avg Temp=30°C Hottest Cell=5	
High Temperature Warning		32 2D 12 00	Max Temp=50°C Avg Temp=25°C Hottest Cell=18	WARNING Temperature warning icon flashes	WARNING Temperature warning icon flashes Max Temp=50°C Avg Temp=25°C Hottest Cell=18	
Critical Temperature		3C 37 12 00	Max Temp=60°C Avg Temp=55°C Hottest Cell=18	SHUTDOWN – STOP CHARGING CRITICAL FAULT	To be checked again	To be checked again
Extreme Temperature		46 3C 25 00	Max Temp=70°C Avg Temp=60°C Hottest Cell=37	SHUTDOWN – STOP CHARGING CRITICAL FAULT	To be checked again	To be checked again
Hottest Cell Change Only		24 20 2A 00	Max Temp=36°C Avg Temp=32°C Hottest Cell=42	SYSTEM OK Hottest Cell updates to 42	SYSTEM OK Hottest Cell updates to 42 Max Temp=36°C Avg Temp=32°C Hottest Cell=42	


CAN BUS Analyzer
File View Tools Setup Help
Rolling Trace
TRACE ID DLC DATA 0 DATA 1 DATA 2 DATA 3 DATA 4 DATA 5 DATA 6 DATA 7 TIME STAMP (sec) TIME DELTA (sec)
Transmit
FORMAT ID DLC DATA 0 DATA 1 DATA 2 DATA 3 DATA 4 DATA 5 DATA 6 DATA 7 PERIOD (msec) REPEAT TRANSMIT
Send
Tool Connected
500 Kbps
Normal Mode
Error Normal
TX ERR: 0
RX ERR: 0
Termination: ON
Trace Active
Logging Inactive
ID in HEX
DATA in HEX
Code_Modification_2 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Adafruit Feather M4 ...
Code_Modification_2.ino
// Update Home Page (page0)
void updateHomePage()
// 1. Check Fault Conditions
bool systemTimeout = (millis() - lastCanMsg) > CAN_TIMEOUT;
bool masterFault = (highTempC >= 60) ||
(socPercent <= 5) ||
(abs(packI) >= 500) ||
systemTimeout;
bool anyWarning = (highTempC >= 50) ||
(packV >= 590) ||
(abs(packI) >= 450) ||
(socPercent <= 15) ||
(cellDelta >= 200);
// 2. Critical Fault Screen
if(masterFault)
static bool lastFault = false;
if(!lastFault)
Output Serial Monitor
100
New Line
115200 baud
DATA: 0x46 0x3C 0x25 0x00
********** DECODED BMS DATA **********
High Temp = 70
Avg Temp = 60
---------- MESSAGE COMPLETE ----------
============= CAN MESSAGE =============
CAN ID: 0x101
DLC: 4
DATA: 0x24 0x20 0x2A 0x00
********** DECODED BMS DATA **********
High Temp = 36
Avg Temp = 32
---------- MESSAGE COMPLETE ----------
Ln 176, Col 1
Adafruit Feather M4 CAN (SAME51) on COM3

Pack Voltage & Current Test Conditions

7.3.3 Pack Voltage and Current Message

The Orion BMS also transmits total pack electrical parameters to the Badenia.

Parameter	Value
CAN ID	0x102
Transmission Rate	10 Hz
Data Length	4 bytes

Signal layout:

Byte	Bits	Signal	Description
0–1	0–15	Pack Voltage	Total accumulator voltag
2–3	16–31	Pack Current	Accumulator current

Pack current is used for overcurrent protection and torque derating.

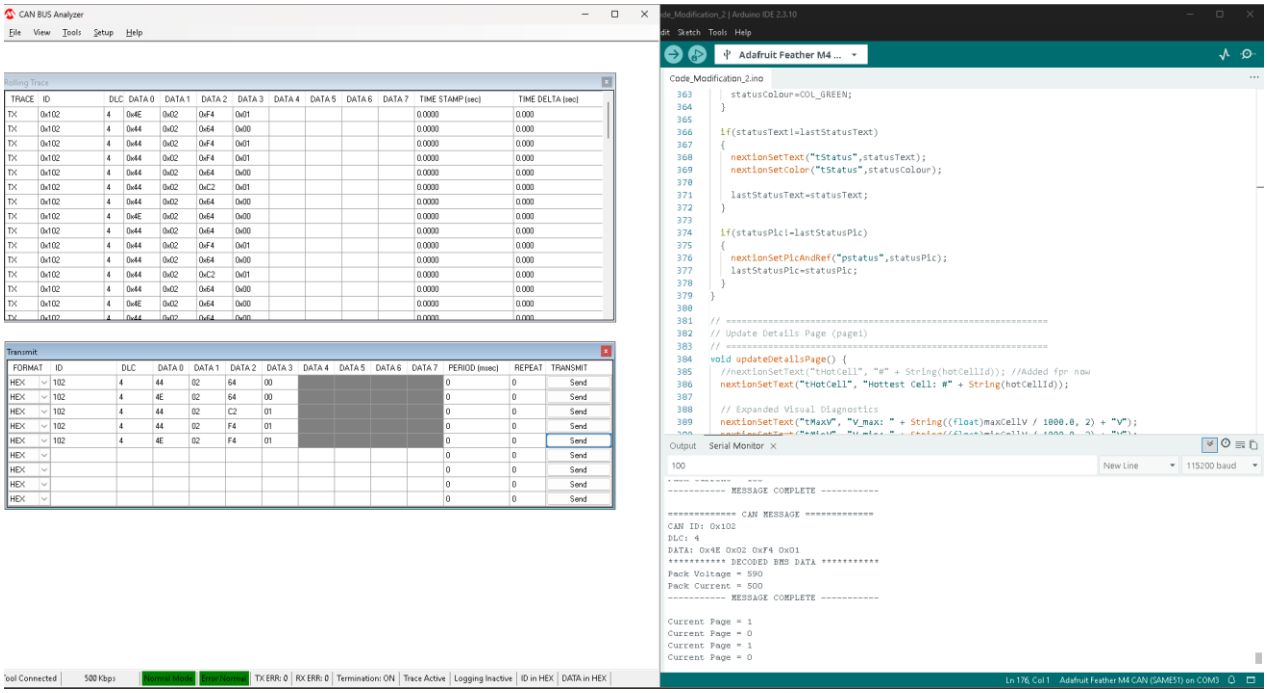
High current draw may indicate excessive electrical load or traction conditions that could overstress the battery system.

Current thresholds are used within the Badenia to apply throttle limiting and protect the accumulator system.

Pack Voltage (V)	Condition	Expected HMI Response
V < 590 V	Normal Operation	SYSTEM OK
V ≥ 590 V	High Voltage Warning	WARNING

Pack Current (A)	Condition	Expected HMI Response
I < 450 A	Normal Operation	SYSTEM OK
450 A ≤ I < 500 A	High Current Warning	WARNING
I ≥ 500 A	Critical Overcurrent	SHUTDOWN

Thermal Cell Condition	Data	Decode Values	Expected HMI	Received HMI	Comments
Normal Operation	44 02 64 00	Voltage = 580 V Current = 100 A	SYSTEM OK No warnings	SYSTEM OK No warnings	Battery shows discharging
High Voltage Warning	4E 02 64 00	Voltage = 590 V Current = 100 A	WARNING High Voltage icon flashes	WARNING High Voltage icon flashes	
High Current Warning	44 02 C2 01	Voltage = 580 V Current = 450 A	WARNING High Current icon flashes	WARNING High Current icon flashes	
Critical Current	44 02 F4 01	Voltage = 580 V Current = 500 A	SHUTDOWN – STOP CHARGING CRITICAL FAULT	To be checked again	To be checked again
High Voltage+High Current	4E 02 F4 01	Voltage = 590 V Current = 500 A	SHUTDOWN – STOP CHARGING Both warning icons active	To be checked again	To be checked again



Note: Charging state validation was not possible as the CAN analyser protocol encodes pack current as an unsigned 16-bit value. Consequently, negative current values required to identify charging are unavailable.

Nextion Display Helper Functions in the program

Function	Purpose
sendEnd()	Terminates every Nextion command with three 0xFF bytes.
nextionSetText()	Updates text displayed by a Nextion component.
nextionSetVal()	Updates the value of numeric components such as the SOC progress bar.
nextionSetColor()	Changes the text colour of a component.
nextionSetBgColor()	Changes the background colour of a component.
nextionSetPic()	Assigns a new image to a Picture component.
nextionRef()	Refreshes a component after modification.
nextionSetPicAndRef()	Updates and refreshes an image in a single function call.
checkNextionEvents()	Detects page changes and updates the active page variable.
socToColor()	Selects the SOC indicator colour based on battery percentage.
updateText()	Sends text updates only when the displayed value changes.
updatePicture()	Sends image updates only when the picture changes, reducing serial communication.

To Do

- Verify the charging scenario to ensure the CAN message is interpreted correctly.
- Verify the system with multiple CAN messages using CAN analyser.
- Connect the system to the BMS and perform final integration tests.

References

[1] B. McEwan, *Control System Handbook: ULM020*. Liverpool, U.K.: University of Liverpool Motorsport, 2026. Unpublished internal technical report

[2] K. Langat, *README: Accumulator Monitoring System Setup and Testing Guide*. Liverpool, U.K.: University of Liverpool Motorsport, 2026. Unpublished internal technical document.